Quality Assurance of a Diagnostic X-ray Machine

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Medical Physics Laboratory II

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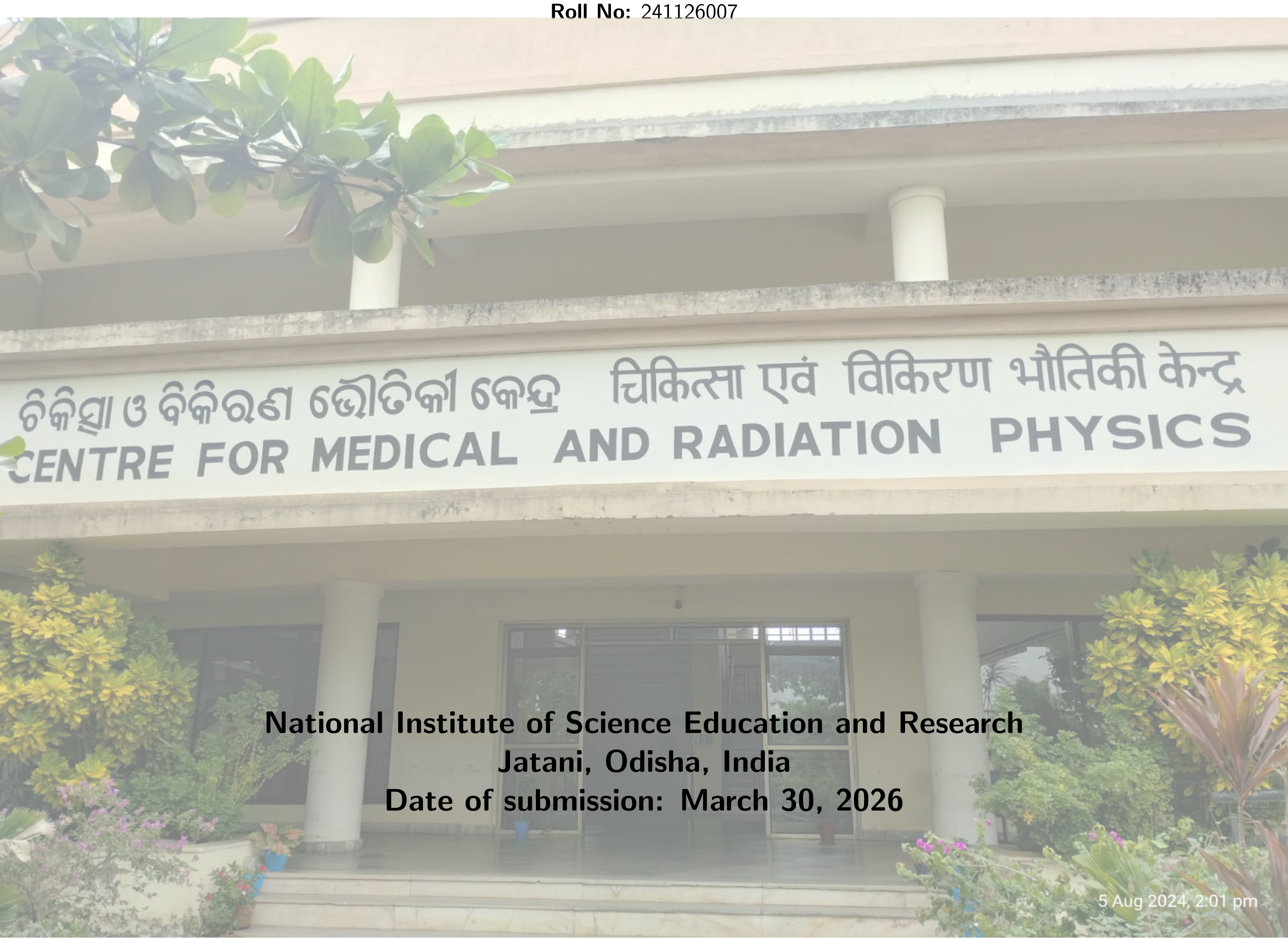

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Contents

1 Objective

The primary objectives of this experiment are:

- To ascertain the Air Kerma Strength (AKS) or the Reference Air Kerma Rate (RAKR) for a High Dose Rate (HDR) Brachytherapy Ir-192 source.
- To calculate the intrinsic timer error of the afterloader system and evaluate its associated percentage error.
- To assess the system's timer linearity along with the end error, determining their respective percentage deviations.

2 Apparatus

The following instruments and materials were utilized to conduct this experiment:

- High Dose Rate (HDR) Brachytherapy Delivery Unit
- Well-type Ionization Chamber Dosimetry System
- Precision Electrometer with required connecting triaxial cables
- Calibrated ambient monitoring devices (Thermometer and Barometer)

3 Theory

3.1 Introduction to Brachytherapy

Brachytherapy is a specialized branch of radiation oncology involving the strategic placement of encapsulated, sealed radioactive sources either directly within or in close proximity to the target tumor volume. Historically derived from the Greek word for "short distance," this technique's core clinical advantage is its ability to conformally deliver an intensely localized radiation dose to a lesion while sharply curtailing the exposure to neighboring healthy tissues. This modality can be implemented as a standalone treatment or as a sequential boost following External Beam Radiotherapy (EBRT).

Prior to clinical application, precise verification of the HDR source's dosimetric strength is mandatory to ensure patient safety and treatment efficacy. The standardized metric for this strength is the Air Kerma Strength (AKS) or the Reference Air Kerma Rate (RAKR). Accurate determination of the AKS/RAKR validates the vendor-supplied source activity and establishes a direct traceable link to primary dosimetric standards. Furthermore, since HDR brachytherapy relies on the rapid delivery of radiation over strictly controlled, short time frames, the absolute accuracy of the afterloader's treatment timer is of critical importance; any temporal discrepancy translates directly into a dosimetric error for the patient.

3.2 Categorization of Brachytherapy

Brachytherapy applications are broadly categorized by three main parameters: dose delivery rate, the anatomical placement of the source, and the temporal duration of the implant.

3.2.1 Classification by Dose Rate

- **Low Dose Rate (LDR):** Operates within a dose rate of 0.4 to 2 Gy/h. The radiation is administered continuously over extended periods (hours to days) using sources like I-125 and Cs-137. It is prevalent in treatments for prostate and certain gynecological malignancies.
- **Medium Dose Rate (MDR):** Ranges from 2 to 12 Gy/h. It serves as an intermediate modality between LDR and HDR but has largely fallen out of modern clinical favor.
- **High Dose Rate (HDR):** Defined by a dose rate exceeding 12 Gy/h. This approach utilizes remote afterloading technology to deliver massive doses in brief, fractionated sessions (minutes). Typical radionuclides include Ir-192 and Co-60. It is a dominant modality for treating cervical, breast, esophageal, and bronchial cancers.

3.2.2 Classification by Source Placement

- **Intracavitary:** The radioactive source is introduced into existing natural anatomical cavities, such as the uterus, vaginal vault, or esophagus. It is a cornerstone technique for cervical cancer.
- **Interstitial:** The sources are surgically implanted directly into the tumor matrix utilizing specialized needles, trocars, or flexible catheters. This technique is routinely applied to prostate, breast, and head and neck tumors.
- **Surface (Mould/Plaque):** Custom-fabricated moulds or applicators hold the sources directly against the external body surface, primarily targeting superficial skin cancers or ocular melanomas.
- **Intraluminal:** The delivery system is threaded through a lumen, such as a bronchus or biliary duct, to treat malignancies arising within hollow tubular organs.

3.2.3 Classification by Implant Duration

- **Temporary Implants:** The radioactive material is positioned for a predetermined timeframe and subsequently retracted. This category encompasses LDR, MDR, and HDR modalities.
- **Permanent Implants:** Tiny, sealed radioactive "seeds" (e.g., I-125) are permanently deposited within the target tissue, where they deliver their total dose over their radioactive lifespan.

3.3 Brachytherapy Radioactive Sources

3.3.1 Encapsulation and Design

To guarantee safety during clinical use and prevent systemic contamination, brachytherapy sources undergo strict encapsulation. The enclosing capsules are typically fabricated from robust metals such as stainless steel, titanium (Ti), or tungsten (W). These specific materials are favored due to their superior mechanical resilience, minimal attenuation of the emitted therapeutic radiation, and excellent biocompatibility.

3.3.2 Photonic Sources

The majority of brachytherapy relies on photon-emitting radionuclides. These sources produce photon spectra with average energies generally spanning from 0.02 MeV to 1.25 MeV. They are further divided by their energy profiles into Low Energy (LE, $\leq$ 50 keV, e.g., ^{125}I, ^{103}Pd) mostly used for permanent seeds, and High Energy (HE, $>$ 50 keV, e.g., ^{192}Ir, ^{60}Co) which demand substantial shielding and are standard for temporary afterloading.

3.3.3 Source Strength Metrics

It is insufficient to rely solely on physical mass or basic construction to predict a source's dosimetric output; therefore, the physical "strength" must be explicitly defined. While historically, strength was denoted in terms of Radium-Equivalent Mass (comparing a source's exposure rate to a standard radium source), modern protocols utilize specific kerma-based quantities.

Air-Kerma Strength (AKS / S_K): The fundamental contemporary metric. It represents the air-kerma rate ($\dot{K}_{air}$) generated by the source in a free-space geometry, theoretically measured at a specific distance r (usually normalized to 1 meter), taking the form:

$$S_K = \dot{K}_{air} \cdot r^2 \tag{1}$$

S_K is typically expressed in units of μGy m^2 h^{-1} (or U).

3.4 Dosimetry via Well-Type Ionization Chamber

The internationally recognized gold standard for the clinical calibration of brachytherapy sources is the re-entrant or well-type ionization chamber. The standard dosimetric assembly includes the vented well-chamber itself, a specialized source-positioning insert (holder), a highly sensitive electrometer, and the necessary cabling.

3.4.1 The Well Chamber Design

Constructed specifically to measure sources approaching a 4π geometry, the chamber features a cylindrical outer housing and a concentric inner well into which the source is inserted. It requires an unsealed, vented design to adapt to atmospheric changes. The device incorporates a central collecting electrode, an outer polarizing electrode, and a guard electrode configured to minimize leakage currents. A crucial characteristic of a well-chamber is its "sweet spot"—a localized axial region yielding maximum and uniform ionization response. The length of this sweet spot must comfortably exceed the physical length of the active source to ensure measurement stability regardless of minor positioning errors.

3.4.2 Measurement Formalism

When utilizing a standard well-chamber, the Reference Air Kerma Rate ($\dot{K}_R$ or RAKR) is derived directly from the measured ionization current:

$$\dot{K}_R = \dot{M}_{corr} \cdot N_K \tag{2}$$

Where $\dot{M}_{corr}$ is the electrometer reading corrected for environmental and operational influence quantities (temperature, pressure, recombination, and polarity), and N_K represents the specific calibration coefficient provided by a Secondary Standard Dosimetry Laboratory (SSDL) for the specific well-chamber, electrometer, and source isotope combination.

3.5 HDR Afterloader Mechanics

The HDR remote afterloader is an advanced, computer-driven apparatus designed to securely store and precisely deploy a high-activity source (such as ^{192}Ir). Using "stepping source" mechanics, a solitary miniaturized source is welded to the end of a drive cable. The system sequentially advances the source through transfer tubes into the applicator, pausing at programmed "dwell positions" for specific "dwell times" to carve out customized dose distributions. The unit houses the source in a dense tungsten safe when idle and incorporates extensive redundant safety mechanisms, including dummy check cables, radiation monitors, and manual retrieval backups.

4 Observation

Prior to taking primary measurements, the source positioning was validated using an external CamScale verification system.

4.1 Determination of Maximum Response Position (Sweet Spot)

The well-type ionization chamber was integrated with the precision electrometer, with the polarity adjusted to match the specifications detailed in the chamber's calibration certificate. A calibrated thermometer and barometer were deployed nearby to log the prevailing ambient temperature and atmospheric pressure.

Following the connection of the HDR source channel to the afterloader system, the ^{192}Ir source was systematically stepped through various dwell positions within the chamber's internal well. The steady-state ionization current was recorded at each specific position to map the chamber's spatial response profile. This procedure identifies the 'sweet spot'—the precise longitudinal position yielding the absolute maximum and most stable dosimetric signal.

Table 1: Ionization current measured across varied dwell positions.

S.No	Dwell Pos. (cm)	Current (nA)	S.No	Dwell Pos. (cm)	Current (nA)
1	131.9	44.033	12	126.4	51.22
2	131.4	45.630	13	125.9	50.79
3	130.9	47.080	14	125.4	50.16
4	130.4	48.340	15	124.9	49.33
5	129.9	49.370	16	124.4	48.25
6	129.4	50.190	17	123.9	46.91
7	128.9	50.815	18	123.4	45.25
8	128.4	51.242	19	122.9	43.26
9	127.9	51.490	20	122.4	40.91
10	127.4	51.560	21	121.9	38.24
11	126.9	51.480			

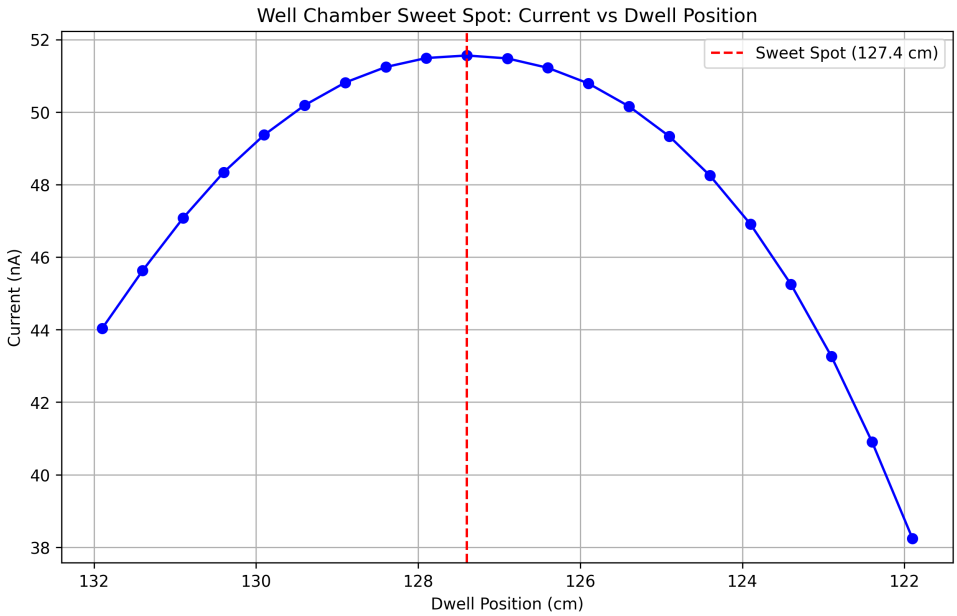

Figure 1: Response curve defining the sweet spot.

Analysis of the response curve clearly indicates a maximum signal intensity at a dwell position of 127.4 cm, generating a peak current of 51.56 nA. This location, mapped to dwell channel number 10, was subsequently utilized for all primary strength measurements.

4.2 Evaluation of Source Strength

With the source localized at the predetermined sweet spot (127.4 cm, channel 10), three independent readings of 20 seconds duration were recorded to ensure statistical reliability.

Table 2: Electrometer readings at the optimal sweet spot.

Dwell Pos.	Polarity (V)	Current (nA)			Avg. Current (nA)
		Trial 1	Trial 2	Trial 3	
10	-300	51.52	51.52	51.52	51.52
10	+150	51.44	51.44	51.44	51.44
10	+300	51.52	51.52	51.52	51.52

4.2.1 Derivation of Correction Factors

To derive the true meter reading, several necessary correction factors must be computed based on operational parameters.

1. Temperature and Pressure Correction (K_{TP}) Given local conditions of $T = 23.5°\text{C}$ and $P = 1010$ mb against reference states $T_0 = 20°\text{C}$ and $P_0 = 1013.2$ mb:

$$K_{TP} = \left(\frac{23.5 + 273.15}{20 + 273.15}\right) \cdot \left(\frac{1013.2}{1010}\right) = 1.015 \tag{3}$$

2. Polarity Correction (K_P) Utilizing readings from opposing polarities (±300V) referenced to the calibration polarity (+300V):

$$K_P = \frac{51.52 + 51.52}{2 \times 51.52} = 1.000 \tag{4}$$

3. Ion Recombination Correction (K_S) Derived via the two-voltage technique (300V vs 150V):

$$K_S = \frac{4}{3} - \frac{51.52}{3 \times 51.44} = 1.0005 \tag{5}$$

4. Electrometer Factor (K_E) Given that the chamber and electrometer were calibrated as a unified pair, $K_E = 1.000$.

The final corrected reading ($\dot{M}_{corr}$) is:

$$\dot{M}_{corr} = 51.52 \text{ nA} \times 1.015 \times 1.000 \times 1.0005 \times 1.000 = 52.32 \text{ nA} \tag{6}$$

4.2.2 RAKR and Source Activity Computation

Using the provided calibration factor $N_K = 4.615 \times 10^5$ Gy h^{-1} A^{-1}:

$$\text{RAKR} = 52.32 \times 10^{-9} \text{ A} \times 4.615 \times 10^5 \text{ Gy h}^{-1} \text{ A}^{-1} = 24.1456 \text{ mGy h}^{-1} \tag{7}$$

Converting this to apparent Activity (A) utilizing the specific air-kerma rate constant ($\Gamma_\delta = 4.07114$ mGy m^2 h^{-1} Ci^{-1}

$$A = \frac{24.1456 \times (1)^2}{4.07114} = 5.931 \text{ Ci} \tag{8}$$

4.3 Timer Error Determination

The intrinsic error associated with source transit and timer delay was evaluated using an interruption methodology over a 60-second exposure. The cumulative charge was compared between a continuous 60s run (R_1) and an equivalent run deliberately halted midway at 30s (R_2).

Table 3: Charge accumulation for timer error test.

Mode	Trial 1 (μC)	Trial 2 (μC)	Average (μC)
Uninterrupted (R_1)	3.103	3.103	3.103
Interrupted (R_2)	3.146	3.146	3.146

The resulting Timer Error is given by:

$$\text{Error} = \frac{(R_2 - R_1) \times t}{2 \times R_1 - R_2} = \frac{(3.146 - 3.103) \times 60}{2(3.103) - 3.146} = 0.8431 \text{ s} \tag{9}$$

4.4 Timer Linearity and End Error Assessment

Timer proportionality was checked across varying set durations relative to a known constant current ($I = 0.05148$ μA, established via a 300s dwell delivering 12.356 μC over a measured 240s window).

Table 4: Linearity check via variable dwell times.

Set Time (s)	Average Charge (μC)	Measured Time (s) = Charge/Current
60	3.103	60.27
120	6.190	120.24
180	9.277	180.20
240	12.365	240.19
300	15.456	300.23

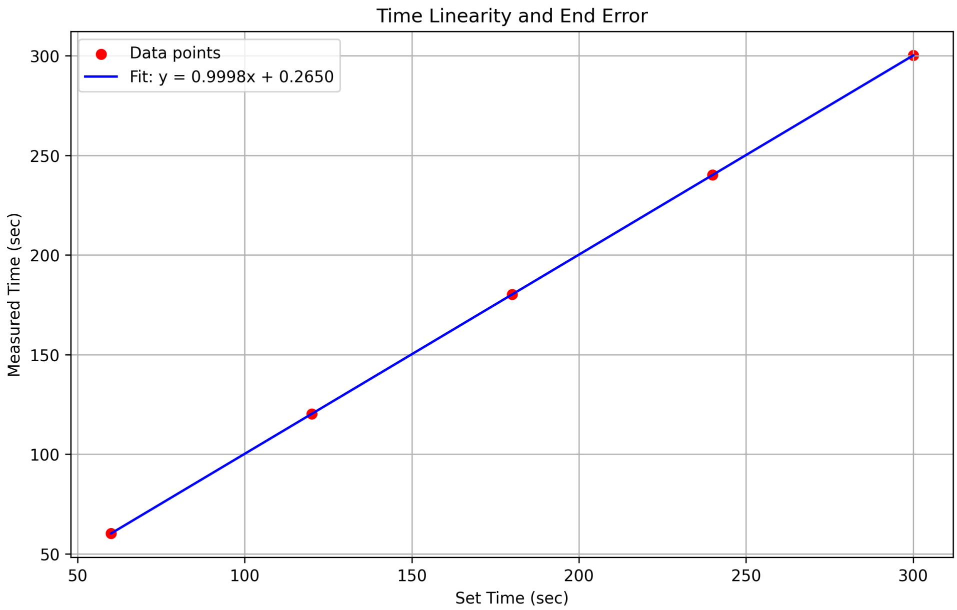

Figure 2: Timer linearity regression analysis.

Applying a linear fit to the measured data yields:

$$\text{Measured Time} = 0.9998 \cdot (\text{Set Time}) + 0.2650 \tag{10}$$

The derived metrics are therefore:

- **End Error (Y-intercept):** 0.2650 s
- **Percentage Time Linearity Error:** $|1 - 0.9998| \times 100\% = 0.02\%$

5 Conclusion

References